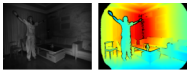
five sensing streams (one training clip: jumping jacks)

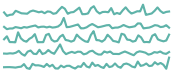
IR + depth video
640 x 480, 16 frames

infrared, depth

thermal video
320 x 240, 16 frames

heat of body and objects

skeleton
17 joints, 3-D

tracked joints, world frame

five inertial units
wrists, ankles, chest

|acceleration|, 5 devices

radar
session key; scene repeated 3x


specialists (accuracy alone, OOF)

YOLO11n crops

person crop, 384 px
MViTv2-S
21.6 MB
0.733

wrist crop, 224 px
MViTv2-S
21.6 MB
0.720

thermal frame, 224 px
MViTv2-S
21.6 MB
0.634

skeleton + IMU
20-member ensemble
18.9 MB
0.618

IMU statistics
ExtraTrees forest
1.6 MB
0.377

fused 0.781

prior-corrected log-linear fusion

$$\sum_i w_i \log p_i$$

coefficients by nested CV

decoded 0.830

ordered transition decoder

activity-script prior, label-free, per recording

recording group, clip order

40
activities

96.4 MB
one file

unseen
subjects

0.877
private, 14th of 307

0/405
changed on 5 GPUs

accuracies: pooled out-of-fold over 2,700 training clips of held-out subjects. private: Kaggle leaderboard, 204 clips of unseen subjects.